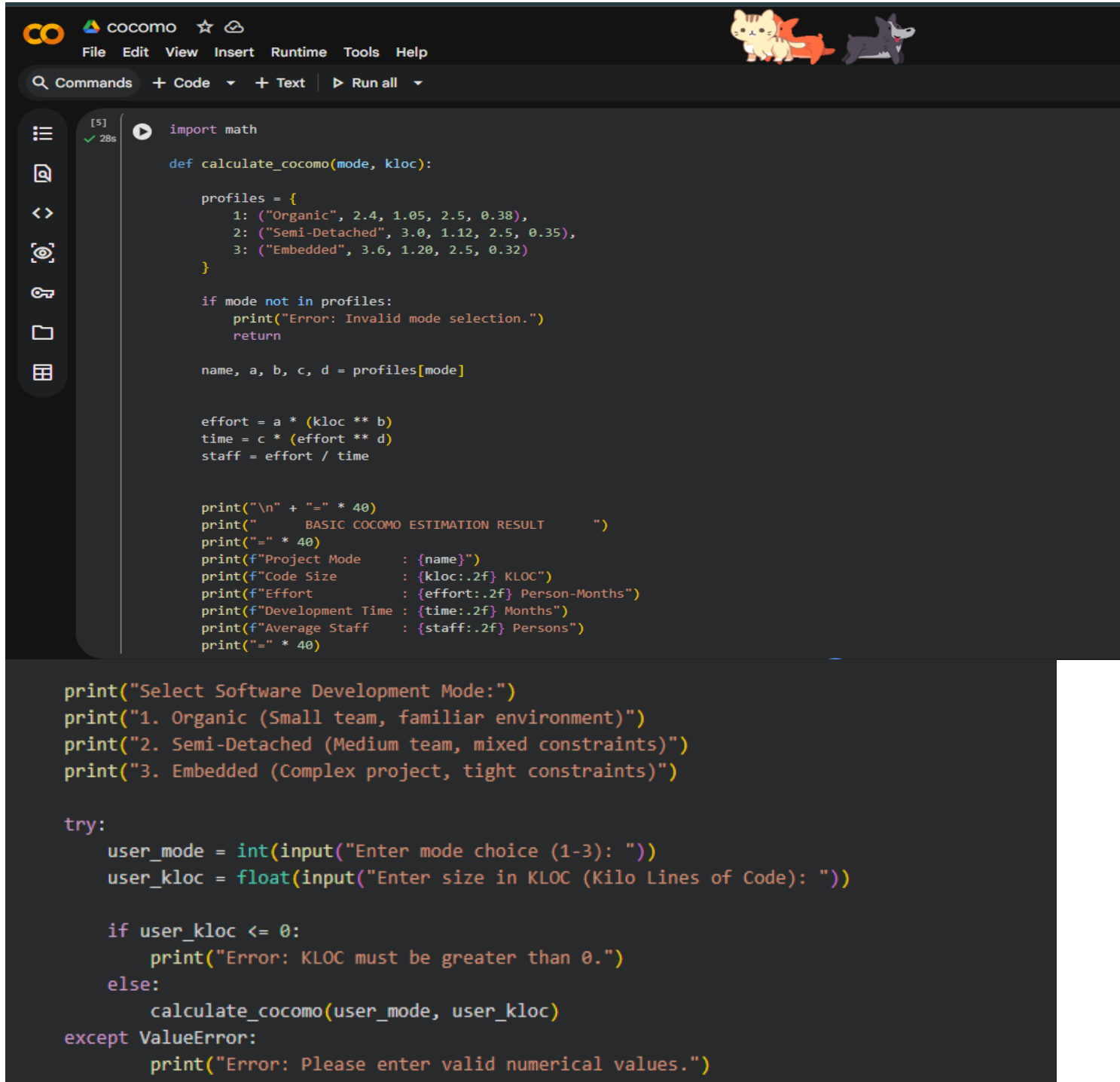


COCOMO MODEL CODE AND OUTPUT



The image shows a code editor interface with a dark theme. At the top, there's a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. Below it is a search bar and a toolbar with icons for 'Commands', 'Code', 'Text', and 'Run all'. The main editor area displays Python code for the COCOMO model. The code defines a function 'calculate_cocomo' that takes 'mode' and 'kloc' as inputs and returns a dictionary with project details. Below the function, there's a main execution block that prompts the user for mode and kloc, validates the input, and prints the results. The output of the program is shown in a separate window at the bottom, displaying the selected mode, code size, effort, development time, and average staff.

```
[5]
✓ 28s
import math

def calculate_cocomo(mode, kloc):

    profiles = {
        1: ("Organic", 2.4, 1.05, 2.5, 0.38),
        2: ("Semi-Detached", 3.0, 1.12, 2.5, 0.35),
        3: ("Embedded", 3.6, 1.20, 2.5, 0.32)
    }

    if mode not in profiles:
        print("Error: Invalid mode selection.")
        return

    name, a, b, c, d = profiles[mode]

    effort = a * (kloc ** b)
    time = c * (effort ** d)
    staff = effort / time

    print("\n" + "=" * 40)
    print("      BASIC COCOMO ESTIMATION RESULT      ")
    print("=" * 40)
    print(f"Project Mode       : {name}")
    print(f"Code Size          : {kloc:.2f} KLOC")
    print(f"Effort              : {effort:.2f} Person-Months")
    print(f"Development Time    : {time:.2f} Months")
    print(f"Average Staff       : {staff:.2f} Persons")
    print("=" * 40)

print("Select Software Development Mode:")
print("1. Organic (Small team, familiar environment)")
print("2. Semi-Detached (Medium team, mixed constraints)")
print("3. Embedded (Complex project, tight constraints)")

try:
    user_mode = int(input("Enter mode choice (1-3): "))
    user_kloc = float(input("Enter size in KLOC (Kilo Lines of Code): "))

    if user_kloc <= 0:
        print("Error: KLOC must be greater than 0.")
    else:
        calculate_cocomo(user_mode, user_kloc)
except ValueError:
    print("Error: Please enter valid numerical values.")
```

Case 1 output

```

... Select Software Development Mode:
  1. Organic (Small team, familiar environment)
  2. Semi-Detached (Medium team, mixed constraints)
  3. Embedded (Complex project, tight constraints)
Enter mode choice (1-3): 1
Enter size in KLOC (Kilo Lines of Code): 2

```

```

=====
          BASIC COCOMO ESTIMATION RESULT
=====
Project Mode      : Organic
Code Size         : 2.00 KLOC
Effort            : 4.97 Person-Months
Development Time  : 4.60 Months
Average Staff     : 1.08 Persons
=====

```

Case 2 output

```

... Select Software Development Mode:
  1. Organic (Small team, familiar environment)
  2. Semi-Detached (Medium team, mixed constraints)
  3. Embedded (Complex project, tight constraints)
Enter mode choice (1-3): 2
Enter size in KLOC (Kilo Lines of Code): 50

```

```

=====
          BASIC COCOMO ESTIMATION RESULT
=====
Project Mode      : Semi-Detached
Code Size         : 50.00 KLOC
Effort            : 239.87 Person-Months
Development Time  : 17.02 Months
Average Staff     : 14.09 Persons
=====

```

Case 3 output

```

... Select Software Development Mode:
  1. Organic (Small team, familiar environment)
  2. Semi-Detached (Medium team, mixed constraints)
  3. Embedded (Complex project, tight constraints)
Enter mode choice (1-3): 3
Enter size in KLOC (Kilo Lines of Code): 100

```

```

=====
          BASIC COCOMO ESTIMATION RESULT
=====
Project Mode      : Embedded
Code Size         : 100.00 KLOC
Effort            : 904.28 Person-Months
Development Time  : 22.08 Months
Average Staff     : 40.96 Persons
=====

```

